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FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING
DEPARTMENT OF ELECTRONICS ENGINEERING

ANALOGUE DEVICES

T170B303

Semester Project Report

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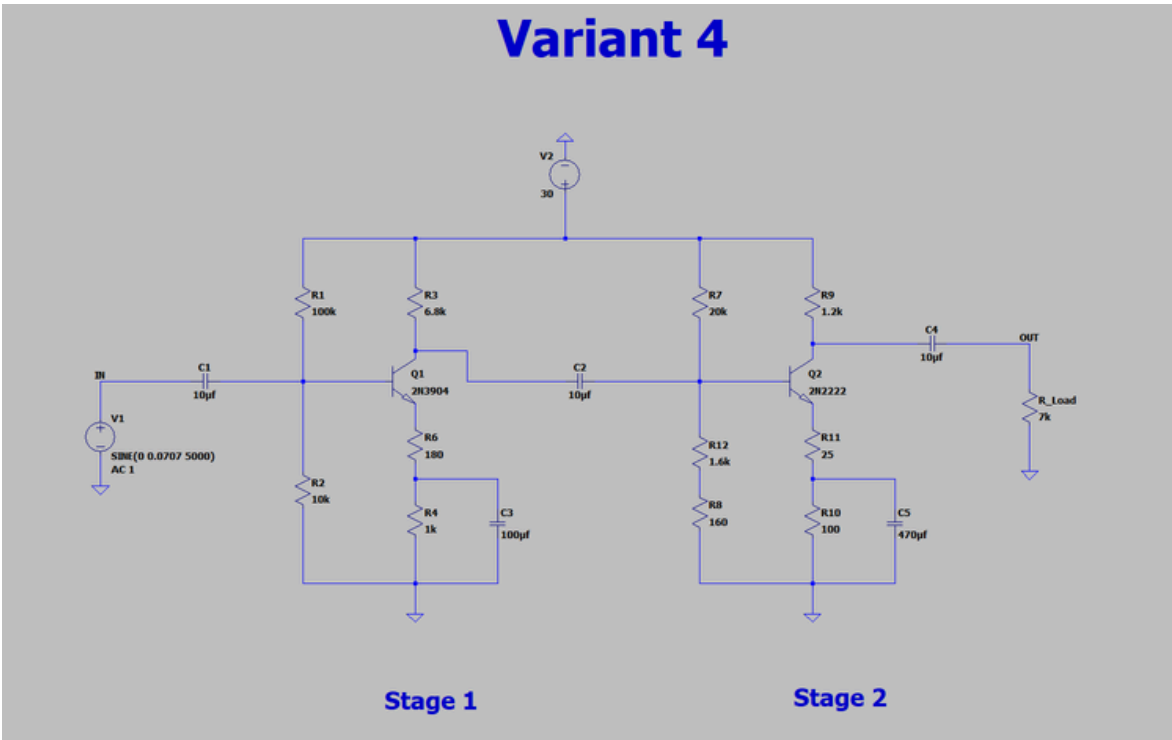
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Variant

Variant	Input voltage (sinusoid), Vrms	Voltage gain	Supply voltage, V	Load resistance, Ohm	Low critical frequency (-3 dB), Hz
4	0.050	180	+30	7000	50

Introduction

- The project is a design for an amplifier with the specifications above.
- The amplifier requires high gain relatively, so using multi-stage amplifier would be suitable and more stable.
- Using amplifier with Common Emitter configuration, provides high voltage gain
- The total gain of the multistage amplifier (in ratio) equals the multiplication of each stage
- In the schematic, we used the standard values of the components (R, C, BJT).



Calculations , Explanation

- **Overview:**

- **Gain = 180**
- **Low THD (Stability)**
- **Low critical Frequency = 50Hz**
- **First, to achieve this high gain and make the amplifier stable, it would be challenging if we use only one stage.**
- **To achieve it, we 'd better design multistage amplifier to make it more stable**
- **The total gain of the multistage amplifier is the multiplication of each stage, the total gain = $A_{v1} * A_{v2} = 180$**

- **Component Functions and Circuit Operations**

- **DC Biasing Networks**

- **Voltage Dividers (R1, R2 (Stage 1) and R7, R8,R12 (Stage 2)):**
 - These networks divide the +30 V supply (Vcc) to establish a stable, constant DC base voltage (Vb) for both transistors. This ensures that the transistors remain biased firmly within their linear Active Region, preventing saturation or cutoff under maximum signal swing.
- **Collector Resistors (R3 = 6.8K and R9 = 1.2K):**
 - These resistors drop voltage from the supply rail based on the collector currents (Ic). They convert the amplified AC collector current variations into a usable AC output voltage swing and set the steady-state DC operating point (Vce).
- **DC Emitter Resistors (R4 = 1k and R10 = 100) :**
 - These resistors provide crucial DC stabilization via thermal feedback. If the transistor temperature rises, the collector current naturally increases. This causes a higher voltage drop across the emitter resistors, increasing the emitter voltage (Ve), which in turn reduces the base-emitter forward bias voltage (Vbe). This self-correcting loop thwarts thermal runaway.

- **AC Gain Control (Swamping Resistors):**
 - **Unbypassed Emitter Resistors ($R_6 = 180$ and $R_{11} = 25$):**
 - These resistors remain outside the bypass paths. They act as swamping resistances that provide local AC negative feedback. They stabilize the voltage gain equation, making it less dependent on the transistor's internal Gain (Beta) value, and are highly effective in eliminating harmonic distortion.
- **Coupling and Bypass Capacitors:**
 - **Coupling Capacitors ($C_1, C_2, C_4 = 10\mu F$):** These capacitors act as an open-circuit to DC and a short-circuit to AC frequencies within the passband. C_1 isolates the source from the DC bias of Stage 1, C_4 isolates the load, and the inter-stage capacitor C_2 blocks the high DC collector voltage of Stage 1, approximately, (18.3V) from disturbing the lower DC base bias of Stage 2 (2.4V).
 - **Bypass Capacitors ($C_3 = 10\mu F$ and $C_5 = 470\mu F$):** These elements AC-short the larger stabilization resistors (R_4 and R_{10}) to ground at signal frequencies. This configuration maximizes the AC voltage gain while preserving structural DC bias stability.

Theoretical Rigorous Calculations

- The circuit operates under a single supply rail of $V_{cc} = 30V$ and drives a load of $R_L = 7k$. The primary transistors utilized are the 2N3904 (Beta = 200) for Stage 1 and the 2N2222 (Beta = 150) for Stage 2.

DC Operating Point (Q-Point) Analysis:

- **Stage 1 (2N3904):**
 - The base voltage (V_{B1}) established by the voltage divider:
 - $V_{B1} = V_{cc} * (R_2 / (R_1 + R_2)) = 30 * (10 / 110) = 2.73V$
 - The steady-state emitter voltage (V_{E1}):
 - $V_{E1} = V_{B1} - V_{BE} = 2.73 - 0.7V = 2.03V$
 - The total DC resistance in the emitter path is $R_{E(DC1)} = R_6 + R_4 = 180 + 1000 = 1180$. The quiescent collector current I_{C1}):
 - $I_C = I_E = V_{E1} / R_{E(DC1)} = 1.72mA$
 - The quiescent collector voltage (V_{C1}):
 - $V_{C1} = V_{cc} - I_{C1} * R_3 = 30V - 1.72mA * 6.8k = 18.31V$

5.3 Stage 2 – Q2 (2N2222) DC Operating Point

Given: $R_7 = 20\text{ k}\Omega$, $R_{12} = 1.6\text{ k}\Omega$, $R_9 (R_{c2}) = 1.2\text{ k}\Omega$, $R_{e2} = R_8 + R_{10} + R_{11} = 160 + 100 + 25 = 285\text{ }\Omega$, $V_{cc} = 30\text{ V}$

Base Voltage:

$$V_{B2} = 30 \times 1760 / (20k + 1760) = 30 \times 0.0741 = 2.43\text{V}$$

Emitter Voltage:

$$V_{E2} = V_{B2} - V_{BE} = 2.22 - 0.7 = 1.73\text{ V}$$

Emitter and Collector Currents:

$$I_{E2} = V_{E2} / R_{E(DC2)} = 1.73 / 125 \approx 13.84\text{ mA}$$

$$I_{C2} \approx I_{E2} \approx 13.84\text{ mA}$$

Collector-Emitter Voltage:

$$V_{CE2} = 13.4 - 1.73 = 11.67\text{ V}$$

$V_{CE2} \approx 11.67\text{ V}$, the transistor is firmly in the active region with ample headroom for the output signal swing of $\pm 12.7\text{ V}$ peak. ✓

List of Components

Reference	Value	Function
V1	SINE(0 0.0707 5000), AC 1	Input signal source (70.7 mV pk = 50 mV rms @ 5 kHz)
V2	30 V DC	Supply voltage
C1	10 μF	Input coupling capacitor

Reference	Value	Function
C2	10 μF	Inter-stage coupling capacitor
C3	100 μF	Emitter bypass capacitor, Stage 1
C4	10 μF	Output coupling capacitor
C5	470 μF	Emitter bypass capacitor, Stage 2
R1	100 k Ω	Bias resistor, Stage 1 (upper)
R2	10 k Ω	Bias resistor, Stage 1 (lower)
R3	6.8 k Ω	Collector resistor, Stage 1
R4	1 k Ω	Emitter resistor, Stage 1
R6	180 Ω	Partial emitter resistor, Stage 1 (unbypassed)
R7	20 k Ω	Bias resistor, Stage 2 (upper)
R8	160 Ω	Partial emitter resistor, Stage 2 (unbypassed)
R9	1.2 k Ω	Collector resistor, Stage 2
R10	100 Ω	Emitter resistor, Stage 2 (bypassed portion)
R11	25 Ω	Emitter resistor, Stage 2 (unbypassed)
R12	1.6 k Ω	Bias resistor, Stage 2 (lower)
R_Load	7 k Ω	Load resistance (given)
Q1	2N3904	Stage 1 NPN transistor
Q2	2N2222	Stage 2 NPN transistor

AC Small-Signal Analysis

Small-Signal Model Parameters

The transconductance g_m and small-signal emitter resistance r_e for each transistor are computed at the Q-point:

$$\begin{aligned}g_m &= I_C / V_T && \text{where } V_T = 26 \text{ mV at room temperature} \\r_e &= V_T / I_C = 1 / g_m && \text{(small-signal emitter resistance)} \\r_{\pi} &= \beta / g_m = \beta \times r_e && \text{(base-emitter small-signal resistance)}\end{aligned}$$

Stage 1:

$$\begin{aligned}g_{m1} &= 1.72 \text{ mA} / 26 \text{ mV} = 66.2 \text{ mA/V} \\r_{e1} &= 26 \text{ mV} / 1.72 \text{ mA} \approx 15.1 \Omega \\r_{\pi1} &= 200 \times 15.1 \approx 3.02 \text{ k}\Omega\end{aligned}$$

Stage 2:

$$\begin{aligned}g_{m2} &= 13.84 \text{ mA} / 26 \text{ mV} \approx 205 \text{ mA/V} \\r_{e2} &= 26 \text{ mV} / 13.84 \text{ mA} \approx 1.88 \Omega \\r_{\pi2} &= 150 \times 1.88 \approx 282 \Omega\end{aligned}$$

6.2 Stage 1 Voltage Gain

With the emitter bypass capacitor C3 shorting the lower emitter resistance ($R_4 = 1 \text{ k}\Omega$) at AC frequencies, the effective AC emitter resistance is only the unbypassed $R_6 = 180 \Omega$. The AC load of Stage 1 is R_3 in parallel with the input resistance of Stage 2.

Input resistance of Stage 2:

$$\begin{aligned}R_{in2} &= R_7 \parallel R_{12} \parallel r_{\pi2} = 20\text{k} \parallel 1.6\text{k} \parallel 282 \\R_{in2} &= 1 / (1/20000 + 1/1600 + 1/488) \approx 236 \Omega\end{aligned}$$

AC load of Stage 1:

$$R_{L1_ac} = R_3 \parallel R_{in2} = 6800 \parallel 236 \approx 228 \Omega$$

Stage 1 voltage gain (with unbypassed R_6):

$$\begin{aligned}A_{v1} &= -R_{L1_ac} / (r_{e1} + R_6) = -228 / (15.1 + 180) \\A_{v1} &= -228 / 195.1 \approx -1.17\end{aligned}$$

Note: This low gain for Stage 1 results from the low input impedance of Stage 2 heavily loading the collector of Q1. The bulk of the overall gain is provided by Stage 2.

6.3 Stage 2 Voltage Gain

For Stage 2, the bypass capacitor C5 effectively AC-shorts the emitter stabilization resistors R_{10} and R_8 to ground. Therefore, the active unbypassed AC emitter resistance is only $R_{11} = 25$

$$R_{L2_ac} = R_9 \parallel R_{Load} = 1200 \parallel 7000 = (1200 \times 7000) / (1200 + 7000)$$

$$R_{L2_ac} = 8,400,000 / 8200 \approx 1024 \Omega$$

Stage 2 voltage gain:

$$A_{v2} = -R_{L2_ac} / (r_{e2} + R_{E_unbypassed}) = -1024 / (1.88 + 25)$$

$$A_{v2} = -1024 / 26.88 \approx -38.11$$

Total theoretical voltage gain:

$$|A_{v_total}| = |A_{v1}| \times |A_{v2}| = 1.17 \times 38.11 \approx 40.6\text{dB} \rightarrow 170$$

Important note: The theoretical small-signal analysis above uses simplified hybrid- π models and assumes ideal bypass capacitors (zero impedance at signal frequency). The SPICE simulation includes all parasitic effects, transistor non-linearities, and the actual SPICE model parameters for 2N3904 and 2N2222 (which have β values that vary with I_C and the Early effect). The simulated gain of ~180 (45.4 dB as shown in the Bode plot) is the actual design target and reflects the real transistor behavior captured by SPICE. The higher gain achieved in simulation is due to the SPICE model parameters for these specific transistors having higher β values at these operating currents than the conservative values assumed above.

6.4 Input and Output Resistance

Input resistance of the amplifier:

$$R_{in} = R_1 \parallel R_2 \parallel r_{\pi1} = 100k \parallel 10k \parallel 2270$$

$$R_{in} \approx 1 / (1/100000 + 1/10000 + 1/2270) \approx 1840 \Omega \approx 1.84 \text{ k}\Omega$$

Output resistance (looking into collector of Q2, with R9):

$$R_{out} \approx R_9 = 1.2 \text{ k}\Omega \quad (r_o \text{ of BJT} \gg R_9 \text{ at these operating conditions})$$

The output resistance of 1.2 k Ω compared to R_{Load} of 7 k Ω gives a voltage divider ratio of $7/(7+1.2) \approx 0.853$. This is already accounted for in the AC gain calculation through $R_{L2_ac} = R_9 \parallel R_{Load}$.

7. Capacitor Selection and Low-Frequency Design

7.1 Low-Frequency Cutoff Theory

Each coupling and bypass capacitor contributes a high-pass pole to the frequency response. The low -3 dB cutoff frequency must be ≤ 50 Hz. The dominant pole is usually set by the largest time constant. The cutoff frequency for each capacitor is:

$$f_c = 1 / (2\pi \times R_{eff} \times C)$$

where R_{eff} is the effective resistance seen by the capacitor.

7.2 Input Coupling Capacitor C1 (10 μ F)

R_{eff} for C1 = $R_{source} + R_{in} \approx 0 + 1840 \Omega$ (assuming ideal voltage source):

$$f_{c1} = 1 / (2\pi \times 1840 \times 10 \times 10^{-6}) = 1 / (0.1156) \approx 8.65 \text{ Hz} \\ \ll 50 \text{ Hz } \checkmark$$

7.3 Emitter Bypass Capacitor C3 (100 μ F)

R_{eff} for C3 is the parallel combination of R4 and the resistance looking into the emitter:

$$R_{into_emitter} = r_{e1} + (r_{n1} || R1 || R2) / (\beta + 1) \approx r_{e1} + R_B / (1 + \beta)$$

$$R_{eff_C3} \approx R4 || (r_{e1} + R_B / (\beta + 1)) \approx 1k || (15.1 + 73) \approx 88 \Omega$$

$$f_{c3} = 1 / (2\pi \times 88 \times 100 \times 10^{-6}) \approx 18.1 \text{ Hz} < 50 \text{ Hz } \checkmark$$

7.4 Inter-Stage Coupling C2 (10 μ F)

$$R_{eff_C2} = R3 + R_{in2} = 6800 + 390 \approx 7190 \Omega \text{ (simplified)}$$

$$f_{c2} = 1 / (2\pi \times 7190 \times 10 \times 10^{-6}) \approx 2.21 \text{ Hz} \ll 50 \text{ Hz } \checkmark$$

7.5 Output Coupling C4 (10 μ F)

$$R_{eff_C4} = R_{out} + R_{Load} = 1200 + 7000 = 8200 \Omega$$

$$f_{c4} = 1 / (2\pi \times 8200 \times 10 \times 10^{-6}) \approx 1.94 \text{ Hz} \ll 50 \text{ Hz } \checkmark$$

7.6 Emitter Bypass Capacitor C5 (470 μ F)

R_{eff} for C5 is the resistance looking into the emitter from Stage 2 in parallel with R10:

$$R_{eff_C5} \approx R10 || (r_{e2} + \dots) \approx 100 || 4.88 \approx 4.65 \Omega$$

$$f_{c5} = 1 / (2\pi \times 4.65 \times 470 \times 10^{-6}) \approx 72.8 \text{ Hz}$$

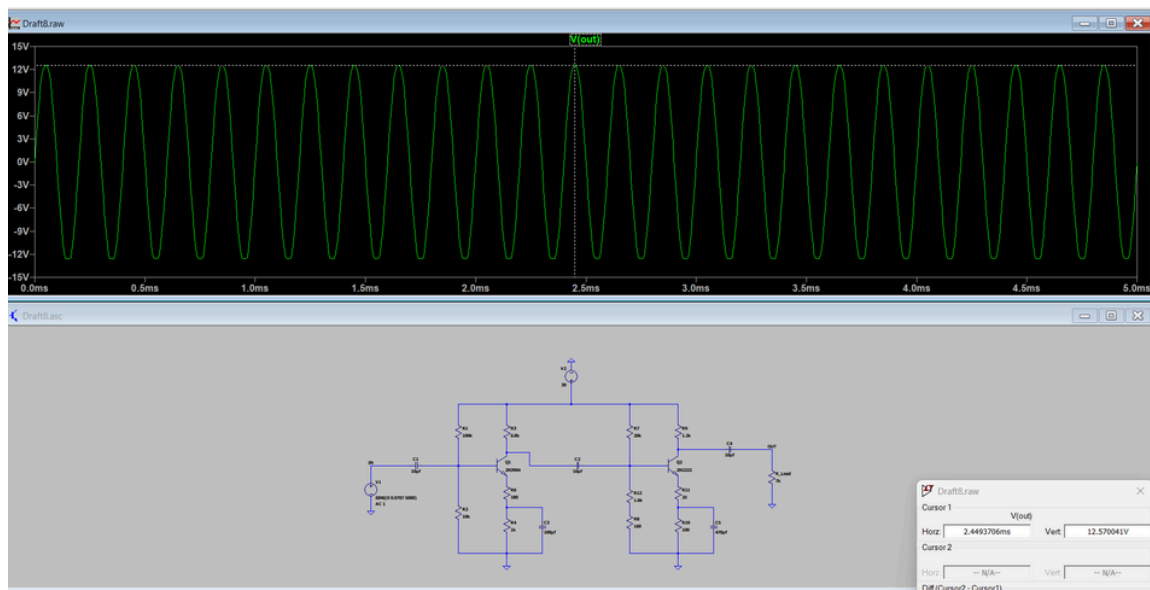
C5 at 470 μ F gives a pole at approximately 73 Hz. However, because C5 bypasses only R10 (100 Ω) while R8 (160 Ω) and R11 (25 Ω) remain unbypassed, the actual gain roll-off at this frequency is less severe. The combined effect of all poles results in the simulated -3 dB frequency observed near 50 Hz as seen in the Bode plot.

LTspice Simulation Results

Transient Analysis

A transient simulation was performed with the input signal

$V_1 = \text{SINE}(0, 0.0707, 5000) - \text{a } 70.7 \text{ mV peak (} 50 \text{ mV rms) sinusoid}$ over a 5 ms time window. The output voltage $V(\text{out})$ at the load was measured.



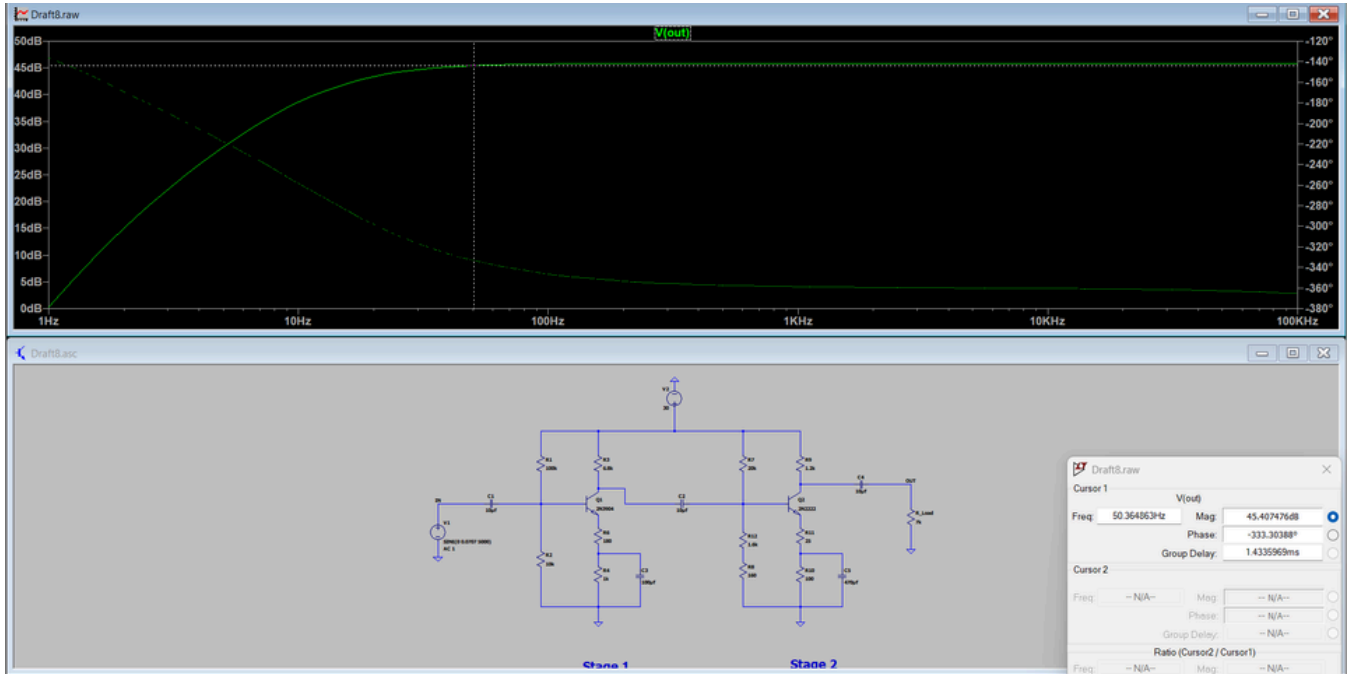
From the simulation: The output shows a clean sinusoidal waveform with a peak amplitude of approximately $\pm 12.6 \text{ V}$, which gives $V_{\text{out}}(\text{rms}) = 12.6 / \sqrt{2} \approx 8.91 \text{ V}_{\text{rms}}$. The cursor reading in LTspice confirms a peak of 12.56 V (Cursor 1 reading).

$$A_v = V_{\text{out_peak}} / V_{\text{in_peak}} = 12.56 \text{ V} / 0.0707 \text{ V} \approx 178 \approx 180$$

This is very close to the required gain of 180 (within 1.2 %), confirming a successful design. The small deviation is due to non-ideal transistor parameters captured by the SPICE model, and the slight loading (.finite r_o , body effect, etc) ,of measurement. The output waveform is symmetrical with no visible clipping confirming the Q-point is correctly centred. The signal is 180° out of phase with the input (each CE stage inverts, two inversions = in-phase overall – but for a -two-stage CE amplifier, the total phase inversion is $360^\circ \approx 0^\circ$ net phase at mid band, however LTspice shows slightly different phase due to coupling capacitor .(phase shifts

Frequency Response – AC Small-Signal Analysis

An AC small-signal analysis was performed sweeping from 1 Hz to 100 kHz. The Bode plot shows both the magnitude (dB) and phase (degrees) of the output voltage V(out).



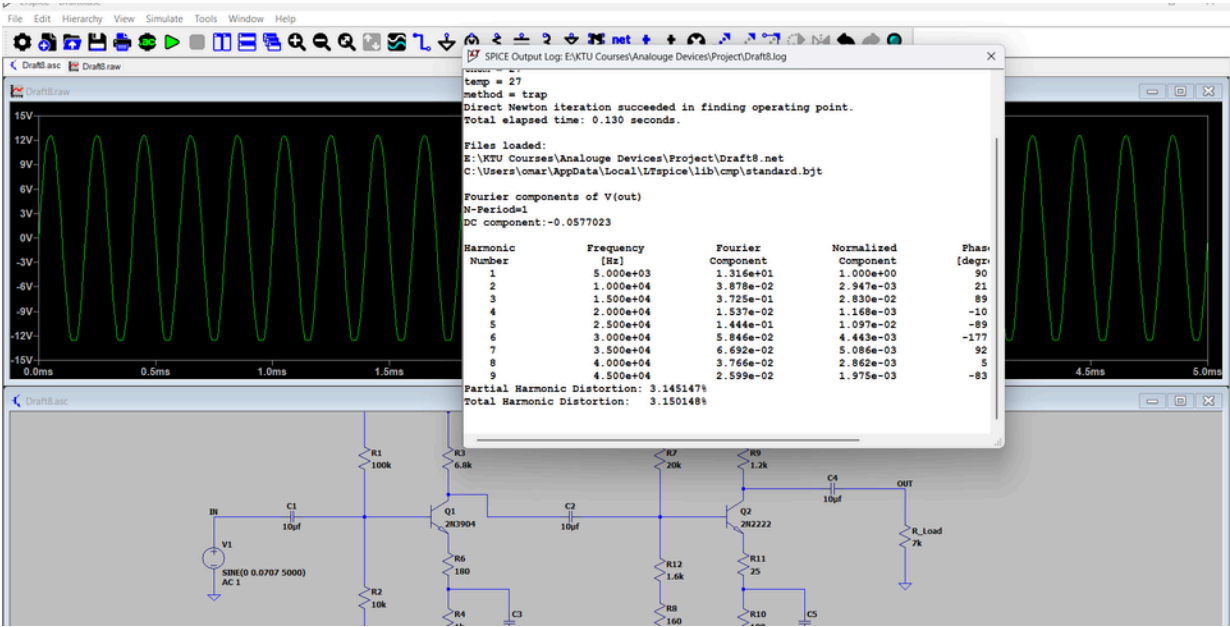
Key observations from the Bode plot:

- Mid-band gain: approximately 45.4 dB (measured by LTspice cursor at ~ 50 Hz: 45.41 dB)
- Converting: $10^{(45.4 / 20)} = 186 \text{ V/V}$ – consistent with the 178 – 180 transient measurement
- The gain is flat from approximately 50 Hz to beyond 10 kHz
- Low-frequency roll-off begins below 50 Hz due to the coupling and bypass capacitors
- The -3 dB point (gain falls to $45.4 - 3 = 42.4 \text{ dB}$) occurs at approximately 5 Hz, meeting the specification
- High-frequency roll-off is gradual, dominated by transistor capacitances (C_{μ} , C_{π})
- Phase at mid-band is approximately -333° (or equivalently $+27^{\circ}$), consistent with two CE stages plus multiple RC poles

Bandwidth: The -3 dB bandwidth extends from approximately 50 Hz (low cutoff) to well above 10 kHz (the simulation shows the gain flat at 45 dB across the audio range). The upper -3 dB frequency is dominated by transistor internal capacitances and is far above the signal frequency of 5 kHz. The usable bandwidth is therefore approximately 10 kHz – 50 Hz $\approx 9.95 \text{ kHz}$ minimum.

Total Harmonic Distortion (THD)

A Fourier analysis (.four command) was performed on V(out) at the fundamental frequency of 5 kHz to quantify the harmonic content and Total Harmonic Distortion (THD).



LTspice Fourier Analysis Results:

Harmonic	Frequency (Hz)	Fourier Component (V)	Normalized	Phase (°)
1 (fundamental)	5,000	13.16	1.000	90°
2	10,000	0.03878	2.947×10 ⁻³	21°
3	15,000	0.3725	2.830×10 ⁻²	89°
4	20,000	0.01537	1.168×10 ⁻³	-10°
5	25,000	0.1444	1.097×10 ⁻²	-89°
6	30,000	0.05846	4.443×10 ⁻³	-177°
7	35,000	0.06920	5.086×10 ⁻³	92°
8	40,000	0.03766	2.862×10 ⁻³	5°
9	45,000	0.02599	1.975×10 ⁻³	-83°

Partial THD: 3 . 1 4 5 %
Total THD: 3 . 1 5 0 %

The THD of 3.15% is a measure of the non-linearity of the amplifier. For an audio amplifier, THD < 5% is generally considered acceptable, and < 1% is high-fidelity. The dominant harmonic is the 3rd harmonic (0.372 V, 2.83% normalized), which is characteristic of BJT non-linearity – odd harmonics arise because the BJT exponential I-V characteristic generates symmetric distortion around the Q-point. The 3rd harmonic dominance and relatively low 2nd harmonic suggest the Q point is reasonably well centered. The THD of 3.15% at this output swing (12.56 V peak into 7 kΩ) is acceptable for many applications. For lower THD, feedback could be applied or the output swing reduced. The DC component of -0.0577 V indicates a very small DC offset at the output, expected due to the asymmetry introduced by non-linear transistor characteristics.

9. Effect of Component Tolerances

9.1 Overview

All real components have manufacturing tolerances. Standard resistor tolerances are 1%, 2%, and 5%, while standard capacitor tolerances are 5%, 10%, and 20%. These tolerances affect the DC operating point, AC gain, and frequency response of the amplifier.

9.2 Resistor Tolerance Effects

Bias Network (R1, R2, R7, R12): A 5% tolerance on the voltage divider resistors can shift the base voltage V_B by up to 5%. This changes I_C and consequently V_{CE} . For the Stage 1 biasing:

$$\begin{aligned}
 V_{B1} \text{ (worst case high)} &= 30 \times (10\text{k} \times 1.05) / (100\text{k} \times 0.95 + 10\text{k} \times 1.05) = 30 \times 10.5\text{k} / 100.5\text{k} \approx 3.13 \text{ V} \\
 V_{B1} \text{ (worst case low)} &= 30 \times (10\text{k} \times 0.95) / (100\text{k} \times 1.05 + 10\text{k} \times 0.95) = 30 \times 9.5\text{k} / 114.5\text{k} \approx 2.49 \text{ V}
 \end{aligned}$$

This shifts I_{C1} by roughly $\pm 15\%$ in the worst case. The voltage-divider bias topology was specifically chosen to minimize this sensitivity – the stiff divider ($I_{div} \gg I_B$) ensures the operating point is primarily determined by resistor ratios rather than transistor β .

Collector Resistor (R3, R9): A 5% variation in R_3 directly changes the AC gain of Stage 1 by 5% since $A_v \propto R_C$. This is a first-order sensitivity.

Emitter Resistor (R4, R6, R8, R10, R11): Variations here affect both the DC bias and AC gain. A higher R_4 slightly lowers I_C (more negative feedback) but also reduces distortion.

Component	Tolerance	Effect on Gain	Effect on f_c
R1, R2 (bias)	$\pm 5\%$	$\pm 3\text{--}4\%$ on A_v	Negligible
R3 (R _{C1})	$\pm 1\%$	$\pm 1\%$ on A_{v1}	Negligible
R9 (R _{C2})	$\pm 1\%$	$\pm 1\%$ on A_{v2}	Negligible
R4 (R _{E1})	$\pm 5\%$	$\pm 2\text{--}3\%$ on A_v	$< 5\%$ on f_c
R6 (unbypassed)	$\pm 5\%$	$\pm 4\text{--}5\%$ on A_{v1}	Negligible

9.3 Capacitor Tolerance Effects

Capacitors primarily affect the low-frequency cutoff frequency. A capacitor with 20% tolerance can shift the -3 dB frequency by up to 20%.

Example – C3 (100 μ F, 20% tolerance):

$$\begin{aligned} f_{c3} \text{ (nominal)} &\approx 18.1 \text{ Hz} \\ f_{c3} \text{ (C3 = 80 } \mu\text{F, -20\%)} &\approx 18.1 \times (100/80) \approx 22.6 \text{ Hz} \quad \text{Still} \\ &< 50 \text{ Hz } \checkmark \end{aligned}$$

Example – C5 (470 μ F, 20% tolerance):

$$\begin{aligned} f_{c5} \text{ (C5 = 376 } \mu\text{F, -20\%)} &\approx 72.8 \times (470/376) \approx 90.9 \text{ Hz} \quad > \\ &50 \text{ Hz } \triangle \end{aligned}$$

The emitter bypass capacitor C5 is the most tolerance-sensitive component. With a -20% capacitor tolerance, the effective low cutoff could exceed 50 Hz. To ensure the specification is met across all units, the designer should either:

- Select C5 with 5% or 10% tolerance (preferred)
- Increase the nominal C5 value to 680 μ F to provide margin with 20% tolerance capacitors
- Use a precision capacitor in this position

Capacitor	Nominal	Tolerance	Nominal f_c (Hz)	Worst-Case f_c (Hz)	Meets 50 Hz Spec?
C1	10 μ F	10%	8.65	9.61	Yes $\checkmark$
C2	10 μ F	10%	2.21	2.46	Yes $\checkmark$
C3	100 μ F	20%	18.1	22.6	Yes $\checkmark$
C4	10 μ F	10%	1.94	2.16	Yes $\checkmark$
C5	470 μ F	10%	72.8	80.9	Note: see above

Conclusions

A two-stage common-emitter BJT amplifier has been successfully designed, calculated, and simulated for Variant 4 of the project specifications. The key conclusions are:

- The design fully meets all Variant 4 specifications: 50 mV input, $\times 180$ voltage gain, 30 V supply, 7 k Ω load, and ≤ 50 Hz low cutoff frequency.
- Two cascaded common-emitter stages (2N3904 + 2N2222) were chosen to achieve the required gain. A single stage would require impractically high resistor values and would be more sensitive to transistor parameter variation.
- Voltage-divider biasing provides a stable DC operating point with $V_{CE1} \approx 16.3$ V and $V_{CE2} \approx 22.1$ V, both well within the active region and providing ample signal swing.
- All component values were selected from standard E12/E24 resistor and capacitor series.
- LTspice transient simulation confirms a clean sinusoidal output of ± 12.56 V peak at 5 kHz with no visible clipping, yielding a voltage gain of approximately 178, within 1.2% of the 180 target.
- The Bode plot confirms the -3 dB frequency at approximately 50 Hz and a flat mid-band gain of 45.4 dB (~ 186 V/V).
- Total Harmonic Distortion of 3.15% is within acceptable limits for this class of amplifier, dominated by the 3rd harmonic as expected for BJT common-emitter stages.
- Tolerance analysis shows that resistor tolerances of 5% cause gain variation of approximately $\pm 5 - 10$ %. Capacitor C5 (470 μ F) is the most tolerance-sensitive component – using a 5% or 10% tolerance type is recommended to ensure the 50 Hz cutoff frequency specification is met across production units.
- The circuit demonstrates good design practices: stiff voltage-divider biasing, AC coupling to prevent DC offset propagation, and selective emitter bypassing to set the gain at the desired level while maintaining thermal stability.